FRAMEWORK FOR AI USE CASE INTEGRATION IN CURRICULUM DELIVERY

Part A: AI Use Case Profile

1. **Use Case ID:** KJS-CES-01
2. **Use Case Title:** Climate Intelligence for Heatwave Monitoring, Prediction, and Early Warning
3. **AI Application Vertical:** Climate, Energy, & Sustainability
4. **Collaborating Organization:** India Meteorological Department (IMD), Mumbai-Pune
5. **Beneficiaries:** State Disaster Management Authorities, Municipal Corporations, Agricultural Agencies, Public Health Departments and Hospitals, Citizens, etc.
6. **Problem Statement:** Heatwaves are among the most impactful extreme weather events, with increasing frequency, intensity, and duration observed across many parts of the world. Prolonged exposure to extreme heat can adversely affect public health, agricultural productivity, ecosystems, and overall socio-economic activities. With climate change, continuous monitoring and early warning of heatwave conditions are essential for improving preparedness and reducing potential impacts. However, heatwave occurrence and severity often vary across regions due to differences in regional weather characteristics and local environmental factors, making accurate forecasting and early warning a challenging task. There is therefore a need for an effective heatwave monitoring and early warning system that can provide timely, reliable, and region-specific information to support decision-making and enhance resilience against extreme heat events.
7. **Background and Context:** Heatwaves have become an increasingly important concern in India due to rising temperatures, rapid urbanization, and changing climatic conditions. In recent years, several parts of the country have experienced prolonged periods of extreme heat, highlighting the need for effective monitoring, forecasting, and early warning mechanisms. To address such events, IMD continuously monitors weather conditions and issues heatwave forecasts and advisories based on established meteorological criteria and observations. However, the diverse climatic characteristics of India and the complex spatio-temporal nature of heatwave development necessitate enhancing heatwave intelligence through more localized monitoring, regional analysis, and advanced data-driven forecasting approaches.

   Recent advances in Artificial Intelligence (AI) and Internet of Things (IoT) technologies provide new opportunities to strengthen heatwave prediction and monitoring capabilities. AI-driven spatio-temporal analysis can help identify complex temperature patterns across different regions and seasons, while IoT-enabled Automated Weather Stations (AWS) can support continuous weather monitoring through localized observations. The integration of these technologies enables a more comprehensive understanding of heatwave behaviour, hotspot evolution, and regional variations in heat stress conditions.

   Along with real-time observations collected through IoT-enabled Automated Weather Stations (AWS), the proposed work will utilize long-term meteorological datasets provided by IMD, comprising historical temperature observations across India. The datasets contain spatial and temporal information, enabling analysis of temperature variations across different climatic regions and seasons. This combination of historical and real-time weather data offers a strong foundation for studying heatwave patterns and developing advanced heatwave intelligence solutions.
8. **Objectives:**

1) To analyze spatio-temporal temperature patterns and heatwave behaviour through region-wise and seasonal analysis of meteorological observations.
2) To develop AI-driven models for region-wise maximum temperature forecasting, heatwave prediction, hotspot identification, and heatwave severity classification.
3) To design and deploy IoT-enabled Automated Weather Stations (AWS) for localized weather monitoring, assessment of local heat conditions, and validation of region-wise heatwave forecasts through integration of ground-based observations with meteorological data.
4) To develop visualization, decision-support, and early warning mechanisms for effective heatwave monitoring and dissemination of heatwave advisories.

9. **Solution Approach:** The proposed solution adopts a data-driven framework for heatwave prediction, monitoring, and early warning by combining historical meteorological observations from IMD with localized weather observations collected through IoT-enabled Automated Weather Stations (AWS). The framework incorporates region-wise analysis, AI-driven forecasting, localized weather monitoring, forecast validation, and stakeholder-specific advisory generation. The overall solution is organized into five interrelated phases, as depicted in Fig. 1 and described below.

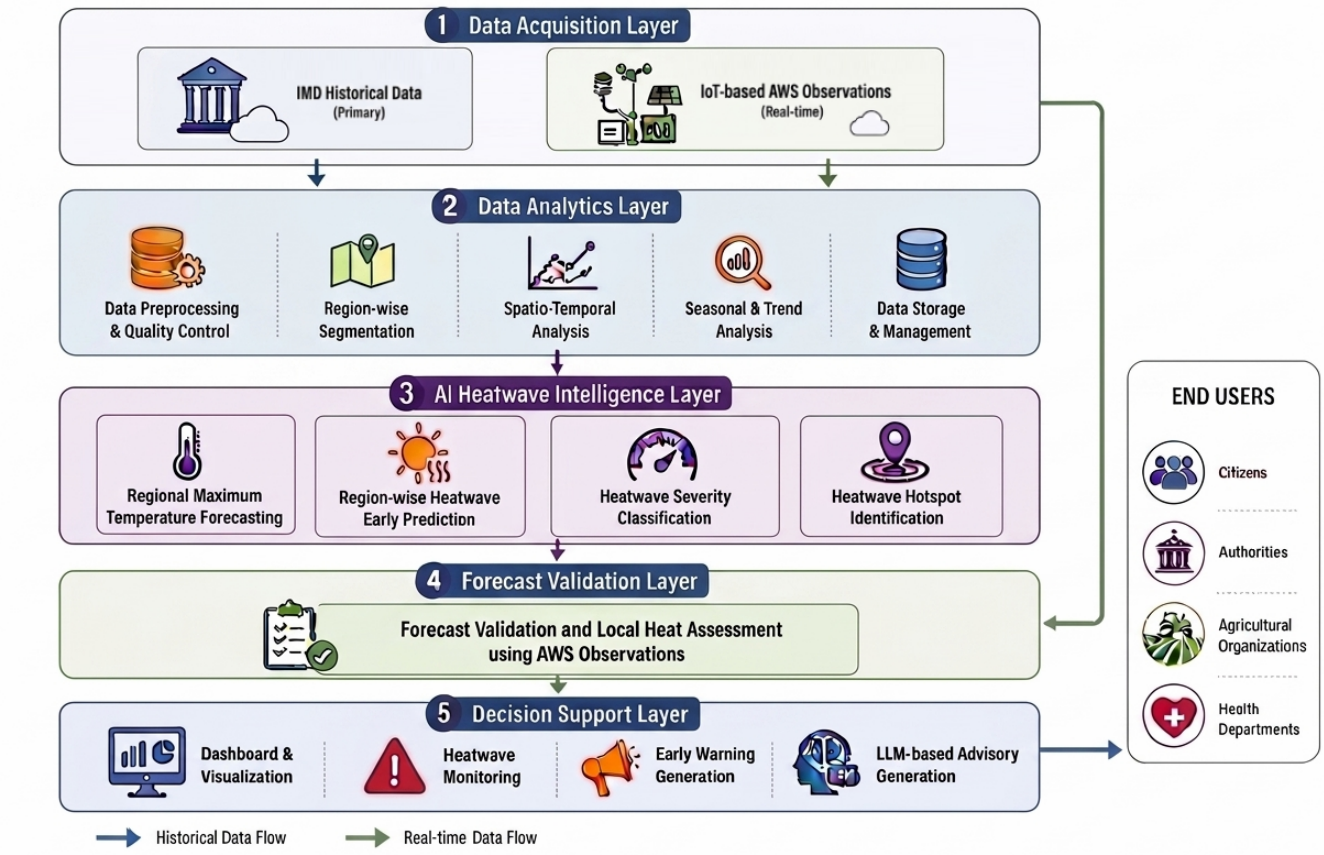


Fig. 1: Conceptual Schema

Phase I: Meteorological Data Acquisition: Historical meteorological observations obtained from IMD will be collected, organized, and stored in a structured repository. Data quality assessment, pre-processing, cleaning, and transformation procedures will be performed to prepare region-wise datasets suitable for subsequent analysis and model development.

Phase II: AI-Driven Heatwave Analytics and Prediction: The processed meteorological data will be subjected to region-wise and seasonal analysis to understand spatio-temporal temperature patterns and heatwave behaviour. AI-driven models will be developed for maximum

temperature forecasting, heatwave prediction, hotspot identification, and heatwave severity classification.

Phase III: IoT-Enabled Weather Monitoring Infrastructure: IoT-enabled Automated Weather Stations (AWS) will be designed and deployed to collect localized weather observations. The infrastructure will support continuous weather monitoring and provide ground-based observations for assessment of local heat conditions.

Phase IV: Forecast Validation and Heatwave Intelligence: Localized observations collected through AWS will be utilized to assess local heat conditions and validate region-wise heatwave forecasts. Combined analysis of meteorological observations, AI-model forecasts, and ground-based measurements will facilitate development of enhanced heatwave intelligence system.

Phase V: Advisory, Early Warning, and Decision Support: A decision-support platform will be developed for visualization of forecasts, heatwave hotspots, severity levels, and monitoring information. Large Language Model (LLM)-assisted advisory generation mechanisms will be incorporated to automatically translate technical forecasts into stakeholder-specific advisories and early warnings for citizens, farmers, health agencies, and local authorities.

10. Operational Challenges:

1) Continuous updating and validation of AI models using newly available observations without degradation in forecasting performance.
2) Selection of suitable AWS deployment locations for effective localized monitoring and representative validation of region-wise heatwave forecasts.
3) Ensuring reliable operation of AWS infrastructure under varying environmental conditions, communication interruptions, and power constraints.
4) Scalability of the proposed system to support multiple regions, larger datasets, additional AWS deployments, and increasing numbers of users.

11. High-Level Workflow:

1) Historical meteorological observations are acquired from IMD, Pune through the designated repository (https://imdpune.gov.in/lrfindex.php) for analysis and model development. The dataset comprises spatial and temporal attributes including Timestamp, Latitude, Longitude, and Maximum Temperature. (IT, CE, AI&DS)
2) Meteorological data is extracted from GRD files containing maximum observed temperature information and undergoes pre-processing, quality assessment, and preparation for spatio-temporal analysis. (IT, CE, AI&DS, CSBS)
3) To capture regional and seasonal variations in heatwave behaviour, the data is segmented into seven IMD-defined regions and four climatic seasons. Region-wise and seasonal analyses are performed to identify temperature patterns, heatwave characteristics, and hotspot behaviour. (IT, CE, AI&DS, CSBS)
4) AI-driven models are developed and utilized for maximum temperature forecasting, heatwave prediction, hotspot identification, and heatwave severity classification. (IT, Computer Engineering, AI&DS, Robotics & AI, CSBS)
5) In parallel, IoT-enabled Automated Weather Stations (AWS) are designed, deployed, and operated to collect localized weather observations from selected locations. (EXTC, CCE, ECE, R AI, MECH, VLSI)
6) Localized AWS observations are used to assess local heat conditions and validate region-wise heatwave forecasts generated by the AI models. (IT, CE, AI&DS)
7) Forecast outputs, monitoring information, validation results, and heatwave intelligence are integrated within a centralized decision-support platform. (IT, CE, AI&DS, CSBS)

8) Visualization modules present forecasts, heatwave hotspots, severity levels, and monitoring information through interactive dashboards and monitoring interfaces. (IT, CE, AI&DS, CSBS)
9) LLM-assisted advisory generation mechanisms convert technical forecasts and heatwave intelligence into stakeholder-specific advisories and early warnings. (IT, CE, AI&DS, CSBS, Robotics & AI)
10) Forecasts, alerts, and advisories are disseminated to citizens, farmers, health agencies, local authorities, and other relevant stakeholders for informed decision-making and preparedness. (IT, CE, AI&DS, CSBS)

12. Expected Outputs:

1) An AI-driven heatwave prediction and early warning system capable of providing region-wise heatwave forecasts and severity assessments (e.g. Fig. 2).

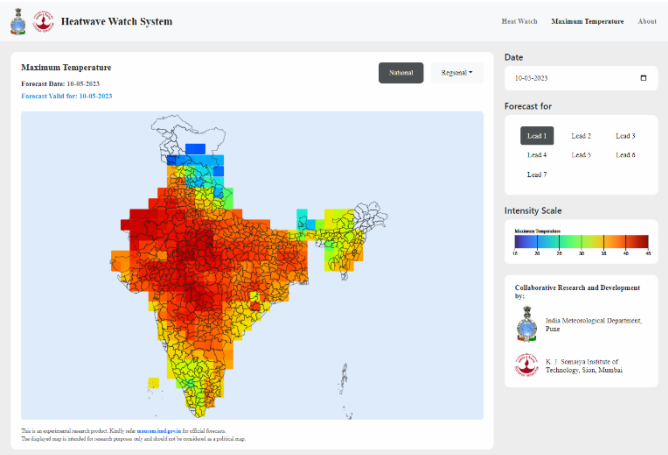


Fig. 2: Temperature Monitoring Software Screen Sample

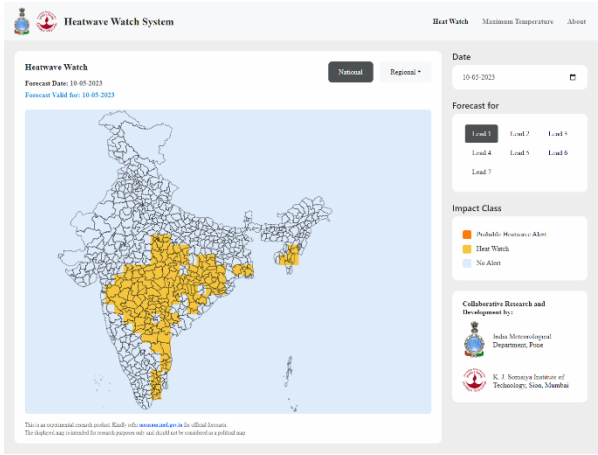


Fig. 3: Heatwave Watch Dashboard Sample

2) An IoT-enabled weather monitoring network comprising Automated Weather Stations (AWS) for localized monitoring and forecast validation.
3) A heatwave intelligence platform for visualization of temperature forecasts, heatwave hotspots, severity levels, and monitoring information (e.g. Fig. 3).
4) An LLM-assisted advisory generation system capable of producing stakeholder-specific heatwave advisories and early warnings.

13. Modern AI Governance and Paradigms:

Area	Relevance	Recommended Consideration
Privacy & Data Protection	Low	Ensure secure storage and controlled access to meteorological observations, AWS data, system logs, and operational information.
AI Security & Trustworthiness	High	Ensure robustness of AI models against erroneous, incomplete, or noisy data and establish mechanisms for forecast validation and performance monitoring.
Responsible AI	High	Promote transparency, fairness, and accountability in heatwave forecasting, advisory generation, and communication of uncertainty in predictions.
Generative AI / LLMs	High	Utilize LLMs for automated generation of stakeholder-specific advisories, alerts, and decision-support information while ensuring factual consistency.

Agentic AI	Low	Explore autonomous monitoring, forecast interpretation, advisory preparation, and alert dissemination workflows to support operational decision-making.
Human-in-the-Loop Decision Making	High	Enable experts, meteorologists, and authorized stakeholders to review, validate, and approve critical forecasts, advisories, and warnings before dissemination.
Regulatory / Governance Considerations	Medium	Ensure compliance with applicable guidelines related to weather information dissemination, AI governance, cybersecurity, and responsible use of forecasting systems.

Part B: Program-wise AI Use Case Mapping

14. Programs Mapped:

Institute: KJSIT		**Institute: KJSSE**	
Information Technology	√	Information Technology	√
Computer Engineering	√	Computer Engineering	√
Artificial Intelligence & Data Science	√	Artificial Intelligence & Data Science	√
Electronics and Telecommunication	√	Electronics & Telecommunication Engineering	√
		Computer & Communication Engineering	√
		Electronics & Computer Engineering	√
		Electronics Engineering (VLSI Design & Technology)	√
		Computer Science & Business Systems	√
		Mechanical Engineering	√
		Robotics & Artificial Intelligence	√

15. Technology Domains Mapped:

Program	**Technology Domain**	**Layer**	**Role in Use Case**	**Inter-Disciplinary Dependencies**
IT, CE	Computing Foundations / Core Competencies / Allied	Data	Storage and management of meteorological, AWS, forecast, and advisory data	Receives meteorological data from IMD & localized observations from AWS (EXTC)
IT, CE, CSBS, AI&DS	Programming and Software Development / Application Development / Computer Software & Systems / Allied	Software	Development of analytics, dashboard, and decision-support applications	Integrates AI outputs with received AWS observations (EXTC)

All	Artificial Intelligence / AI & ML / Data Science & Analytics / Knowledge Based Systems / Intelligent Systems / Allied	AI	Heatwave prediction, temperature forecasting, hotspot identification, and advisory generation	Utilizes pre-processed meteorological data (IT/CE) and AWS observations (EXTC)
EXTC, VLSI, CCE, ECE	Embedded Systems / Electronic Devices & Linear Integrated Circuits / Sensor Technology / Circuits and Systems / Allied	IoT / Hardware	Design and development of IoT-enabled AWS, sensor interfacing, data acquisition, and edge monitoring systems	Provides localized weather observations for analytics, forecasting, and validation modules (IT, CE, CSBS, AI&DS)
	System Software Engineering / Programming and Software Development / Systems and Software / Allied	Software + IoT	Development of firmware, device management, communication protocols, and AWS operational software	Interfaces IoT hardware with software platforms (IT, CE, CSBS, AI&DS)
RAI, Mech	Manufacturing / Product Design / Emerging Technologies	Mechanical / Hardware	Design of AWS enclosures, mounting structures, environmental protection mechanisms, and field deployment systems	Supports AWS infrastructure development (EXTC)

Part C: Course-level AI Use Case Mapping

16. Course-level AI Use Case Mapping:M

Institute: KJSSE Program: Computer Engineering			
Course	**Contribution to Use Case**	**Relevant Topics from Syllabus**	**Potential Modes of Integration**
Database Management Systems	Design and implementation of meteorological database	All modules	Classroom case study, laboratory experiments, IA tool
Web Development Laboratory			
Software Engineering			

Computer Graphics (Elective)			
Modern Data Architectures (Elective)			
Software Testing and QualityAssurance (Elective)	Design, Develop and execute Test cases for the given use case..	Entire Syllabus (to be specific Module 2 and 3)	Can be used in laboratory and one of the IA tools
Soft Computing			

Part D: Inter-disciplinary Faculty Committee

17. **Faculty Owner(s):** Dr. Radhika Kotecha, Professor and Head, Department of Information Technology, K J Somaiya Institute of Technology

18. **Program-level Integration Coordinators:**

Institute	Program	Coordinator
KJSIT	e.g. IT	Faculty Name & Designation
KJSSE	e.g. Mechanical	Faculty Name & Designation

19. **Course-wise Integration Coordinators:**

Institute	Program	Course	Course In-charge
KJSIT	e.g. IT		Faculty Name & Designation
KJSSE	e.g. Mechanical		Faculty Name & Designation

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